# Akoizo Original Practice - Form 2 - Answer key

### Division B | Codebusters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

#### 1. (2 points) What is the plaintext?

Unit: Atbash and Caesar shifts.
Worked-case data: Cipher: Caesar with forward encryption shift 3. Ciphertext: FDW. Alphabet: A=0 through Z=25.
Separate investigation observations:
Cipher: Caesar with forward encryption shift 3.
Ciphertext: FDW.
Alphabet: A=0 through Z=25.

A. CAT

B. DOG

C. XZD

D. IGZ

**Correct: A**

#### 2. (2 points) How many binary positions encode one letter here?

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

A. 3

B. 26

C. 5

D. 2

**Correct: C**

### 3. (2 points) What plaintext results when CTO is the left column?

Unit: Columnar transposition and homophonic evidence.
Worked-case data: Rectangle: Six letters in two columns. Ciphertext pieces: CTO and ATO. Clue: The plaintext starts with CAT.
Separate investigation observations:
Rectangle: Six letters in two columns.
Ciphertext pieces: CTO and ATO.
Clue: The plaintext starts with CAT.

A. CATTWO

B. CTOATO

C. CATTOO

D. ATOC TO

**Correct: C**

### 4. (2 points) What is removed in a patristocrat?

Unit: Monoalphabetic substitution and language evidence.
Worked-case data: A ciphertext word ABBC is proposed to mean MEET or TEAM. Which fits the equality pattern?

A. All vowels

B. Normal word boundaries

C. All repeated letters

D. The substitution key space

**Correct: B**

### 5. (2 points) What should follow an arithmetic-error review?

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

A. A new problem exercising the corrected skill

B. Memorize the old answer only

C. Skip all verification

D. Change every cipher family

**Correct: A**

**6. (2 points) What begins a deduplicated BALLOON keyword alphabet?**

Unit: Monoalphabetic substitution and language evidence.
Worked-case data: A ciphertext word ABBC is proposed to mean MEET or TEAM. Which fits the equality pattern?

A. AB LNO

B. BONLA

C. BALON

D. BALLOON

**Correct: C**

**7. (2 points) Why can a numerically true sum still be invalid?**

Unit: Xenocrypts and cryptarithms.
Worked-case data: Verify the candidate SEND = 9567, MORE = 1085, MONEY = 10652.

A. It contains a zero internally

B. It uses eight letters

C. Different letters may share a forbidden digit

D. It has any carry

**Correct: C**

**8. (4 points) Encode ZOO with key +3.**

answer: CRR: Z wraps to C and O maps to R twice. (4 points).

**9. (6 points) Solve this case and explain your reasoning: Nihilist ciphertext: 65. Key coordinate: 23. Square entry: Row 4, column 2 contains R.**

reasoning: Subtract whole coordinate numbers: 65 - 23 = 42.
Interpret 42 as row 4, column 2.
Read R and check 42 + 23 = 65. (3 points).

conclusion: R (3 points).

**10. (4 points) Compute A's expected points per minute.**

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

answer: 100 x 0.9 / 3 = 30 expected points/minute under the estimate. (4 points).

**11. (4 points) Why does the most frequent symbol not prove a mapping to E?**

Unit: Monoalphabetic substitution and language evidence.
Worked-case data: A ciphertext word ABBC is proposed to mean MEET or TEAM. Which fits the equality pattern?

answer: Short samples and topic-specific vocabulary can depart from average frequencies; independent constraints are required. (4 points).

**12. (4 points) Why is the heuristic uncertain?**

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

answer: Solve probabilities and times are estimates; dependencies, scoring rules, and later clues can change priorities. (4 points).

**13. (4 points) A decoded word is misspelled in an erristocrat. What should be checked first?**

answer: Verify consistent substitution and re-encryption before treating the spelling as an intentional source error. (4 points).

**14. (4 points) How should a guessed Spanish DE be tested?**

Unit: Xenocrypts and cryptarithms.
Worked-case data: Verify the candidate SEND = 9567, MORE = 1085, MONEY = 10652.

answer: Propagate both letter mappings to all occurrences and check grammar and substitution consistency. (4 points).

**15. (4 points) Design a useful entry for an error log.**

Unit: Cryptography foundations and the solving workflow.
Worked-case data: A = 0 and a Caesar key of +3 are used to encode CAT. Decode FDW and verify.

answer: Record the cipher, mistaken assumption, corrected transformation, and a short re-encrypted example that verifies the correction. (4 points).

**16. (2 points) Define “Crib” in the context of Codebusters.**

answer: Known plaintext used to constrain a solution. (2 points).

**17. (4 points) Design a short interleaved drill.**

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

answer: Mix a shift, a pattern substitution, and a coordinate puzzle; identify the family and conventions before solving and verify each answer. (4 points).

**18. (2 points) Define “Plaintext” in the context of Codebusters.**

answer: The original readable message. (2 points).